

Project Planning

Project: AI Study Material Generator

Team Name: CSE4104-7C-T03

Project Overview

The AI Study Material Generator is an AI-powered educational platform designed to help students automatically generate study resources such as summaries, MCQs, flashcards, quizzes, and important questions from uploaded study materials.

The system will use Artificial Intelligence and Natural Language Processing (NLP) to analyze educational documents and create personalized learning content.

Project Goals

- Develop an intelligent study assistant system
 - Reduce manual effort in preparing study materials
 - Improve learning efficiency through AI-generated content
 - Provide automated quizzes and revision materials
 - Build a user-friendly educational platform
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Core Modules Planning

Module	Description
Authentication Module	User registration and login system
File Upload Module	Upload PDFs, notes, and study materials
AI Processing Module	Analyze uploaded content using AI
Summary Generator	Generate smart summaries
MCQ Generator	Create automatic quiz questions
Flashcard System	Generate revision flashcards
Dashboard Module	Manage uploaded and generated materials
AI Chatbot	Answer student queries

Development Phases

Phase 1 — Planning & Research

- Finalize project idea
- Research similar AI systems
- Prepare proposal and documentation
- Create GitHub repository

Phase 2 — UI/UX Design

- Design homepage
- Create dashboard design
- Design upload interface
- Prepare wireframes

Phase 3 — Frontend Development

- Build user interface using React.js
- Create authentication pages
- Develop dashboard and upload system

Phase 4 — Backend Development

- Setup Node.js server
- Create APIs
- Connect database
- Handle file uploads

Phase 5 — AI Integration

- Integrate OpenAI API
- Implement NLP processing
- Generate summaries and MCQs
- Build chatbot system

Phase 6 — Testing

- Bug fixing
- Feature testing
- Performance optimization
- UI improvements

Phase 7 — Deployment & Final Submission

- Deploy project online
 - Prepare presentation
 - Complete documentation
 - Final project submission
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Technology Planning

Frontend

- HTML
- CSS
- JavaScript
- React.js

Backend

- Node.js
- Express.js

Database

- MongoDB

AI Technologies

- OpenAI API

Deployment

- Vercel
 - Render
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Team Task Distribution

Member Role	Responsibilities
Team Leader	Project management, GitHub management, backend supervision
Frontend Developer	UI/UX and frontend development
Backend Developer	API and database development
AI Developer	AI integration and NLP features

Expected Challenges

- AI response accuracy
 - Processing large PDF files
 - Integrating AI APIs efficiently
 - Managing project timeline
 - Database optimization
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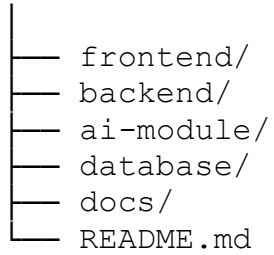
Expected Outcome

The final system will:

- Generate study materials automatically
 - Improve student productivity
 - Save study preparation time
 - Provide intelligent educational assistance
 - Support personalized learning experiences
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GitHub Repository Structure

AI-Study-Material-Generator/



```
├── frontend/
├── backend/
├── ai-module/
├── database/
├── docs/
└── README.md
```

Future Improvements

- Mobile application support
- Voice-based AI assistant
- Multi-language support
- AI-based learning recommendations
- Student performance analytic